Title: *Evaluating Template-Based Stellar Spectral Classification Using SDSS Spectra*

Abstract:

This paper presents a Python-based pipeline for automated stellar spectral classification using spectroscopic data from the Sloan Digital Sky Survey (SDSS). By retrieving template spectra and utilizing normalized feature-masked cross-correlation against absorption lines, the program classifies unknown stellar spectra within the Morgan-Keenan (MK) classification system. Testing procedures and dataset retrieval strategies are evaluated, alongside an analysis of selection biases and proposed algorithmic improvements for future development.

Introduction:

Historically, stars have been categorized into classifications based on their physical and thermal properties. The standard framework used in modern astrophysics is the Morgan–Keenan (MK) classification system, which integrates the Harvard spectral classification (based on surface temperature) with luminosity class. The primary spectral types are O, B, A, F, G, K, and M, ranging from the hottest (O) to the coolest (M). Each class is further subdivided decimally from 0 to 9, where 0 represents the hottest subtype and 9 the coolest (for reference, our Sun is classified as a G2 star). While luminosity classes are denoted by Roman numerals, this project focuses specifically on temperature/spectral class determination.

Stellar classification can be determined by analyzing stellar spectra, where the overall continuum shape and specific absorption lines serve as strong reference points. The objective of this project is to develop a Python program that takes in an observed stellar spectrum and automatically outputs its stellar spectral type and decimal subclass within the MK classification system by comparing it against reference templates of known stars.

Methodology and Data Collection:

All spectroscopic data for this project were retrieved from the Sloan Digital Sky Survey (SDSS) through SQL queries specifying plate and fiber identifiers (plate and fiberID).

Template Generation

Initially, template data was retrieved directly from SDSS template documentation; however, these templates lacked diversity, subtype specificity, and accuracy. To resolve this, a template generation system was built using targeted SQL queries that retrieve one spectrum for each stellar subtype.

One key parameter present in this query is snMedian, which evaluates the median signal-to-noise ratio (SNR) across the spectrum. Higher SNR values correspond to cleaner spectra and absorption line data. Sorting by descending snMedian ensures that each reference template represents the high quality data, minimizing noise that could disrupt the comparison phase.

Normalization and Matching

Because observed stellar intensities vary significantly due to distance and apparent magnitude, it is impossible to directly compare stellar spectra. To address this, the program first normalizes both the observed spectrum and the reference templates around a baseline value of 1.

Once normalized, the program computes a "score" for how well each template fits the spectrum, and the template with the lowest score is selected as the best match. This score is determined through a three step comparison process:

1. Wavelength Alignment: Because observed and template spectra can have slightly different wavelength intervals, the program interpolates the template onto the observed star's wavelength grid so they can be properly compared
2. Smoothing and Feature Masking: To prevent background noise from throwing off the results, the program smooths the template flux and applies a feature mask. This isolates only the most notable absorption lines rather than comparing the entire spectrum.
3. Cross-correlation scoring: The script calculates the similarity of these isolated absorption features using a cross-correlation calculation that assigns a score to each template. Lower scores indicate closer alignments, so the template that produces the lowest score is selected as the final classification match.

To help with verification and debugging, the program includes plotting mechanisms that are capable of rendering both unprocessed (Figure 1) and normalized spectra. Normalized spectra can also be overlaid directly onto any given template (Figures 2, 3), making visual stellar classification significantly easier.

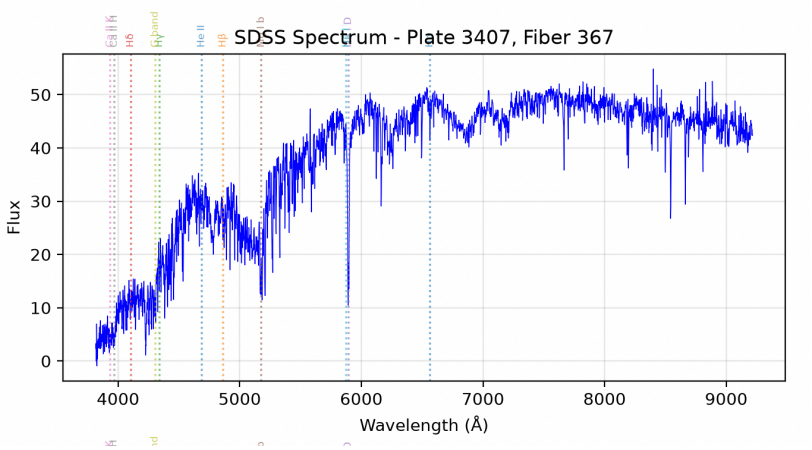


***Figure 1**: An unfiltered F5 star spectrum*

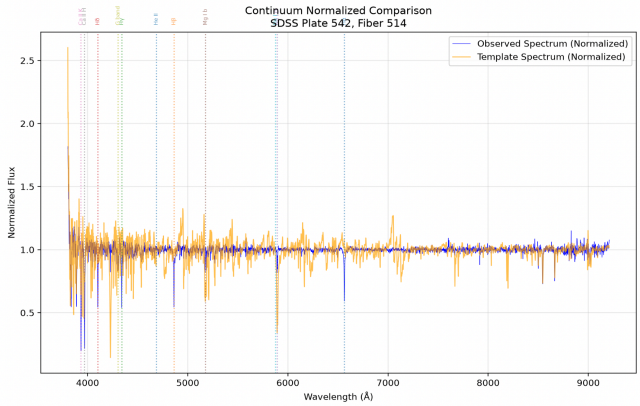


***Figure 2**: A normalized F5 spectrum overlaid with an incorrect M3 template*

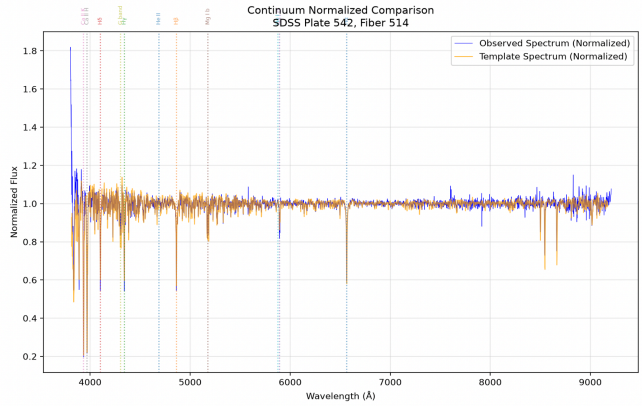


***Figure 3**: A normalized F5 spectrum correctly matched and overlaid with an F5 template*

The program also supports processing on a larger scale through coordinate-based searches using Right Ascension (ra), Declination (dec), and a search radius. This is much more useful for large-scale classification as it does not require individually entering the IDs of each star.

Results and Discussion:

Evaluating the accuracy of the classifier presented notable challenges. The initial testing pipeline selected test spectra strictly by descending snMedian to minimize noise. However, this created a harsh selection bias, as the SNR requirement resulted in there being an overwhelming majority of F-type stars among the stars selected. Since F-type stars comprise only a small fraction (~3%) of main sequence stars, the testing process had to be reevaluated.

To resolve this sampling bias, the testing function was updated to test subtypes independently to calculate accuracy specific to individual star types. This allowed for target queries (e.g., specific subtypes such as M7 or broader groupings like M) to generate a more balanced accuracy assessment across the entire spectral sequence.

Outputs were compared to official SDSS classifications across sample subsets of 200 stars per spectral class:

Spectral Type	Class Accuracy	Subclass Accuracy
O	1.0%	0.5%
B	59.5%	15.0%
A	46.1%	9.3%
F	81.2%	43.7%
G	9.9%	1.0%
K	74.2%	29.9%
M	60.5%	17.9%

(Note: "Class Accuracy" reflects correct identification of the spectral class letter, while "Subclass Accuracy" measures correct identification of both its primary spectral class and decimal subtype).

While classes such as F (81.2% letter accuracy) and K (74.2%) demonstrated strong performance, classes like O (1.0%) and G (9.9%) exhibited high misclassification rates. Several factors contribute to these discrepancies:

1. Relying on a single reference spectrum for each subclass does not account for the diversity that exists within each classification
2. Some spectral features may be more difficult for the matching function to distinguish
3. The class accuracy system treats an F9 star classified as a G0 star as a complete failure, even though F9 and G0 are technically adjacent and far more similar than F9 and F0.

To address these limitations, future versions of this program could implement the following improvements:

1. Querying multiple stars per decimal subclass to generate median templates that provide a more encompassing comparison for each spectrum
2. Developing an additional accuracy metric that quantifies classifications that are relatively close (e.g., F9 and G0) that may not share the same letter
3. Adding luminosity as another dimension to be included in classification

References:

Gray, R. O., & Corbally, C. J. (2014). An expert computer program for classifying stars on the MK spectral classification system. *The Astronomical Journal, 147*, Article 80. https://doi.org/10.1088/0004-6256/147/4/80

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Imamura, J. (n.d.). *Morgan-Keenan (MK) spectral classification*. University of Oregon. University of Oregon